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# Adaptive event-triggered $H_\infty$ consensus secondary control for islanded microgrids under multiple time delays

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ABSTRACT

In this paper, an adaptive event-triggered $H_\infty$ consensus control scheme is proposed for the secondary layer of AC MGs. The controller design problem of topological communication network with external disturbance, multiple delay, and communication burden is considered. Therefore, using the Lyapunov-Krasovskii theory, we obtain sufficient conditions for the robust consensus convergence of multi-agents under event-triggered and multiple time delays. Because traditional event-triggered often use fixed thresholds, it is challenging to balance reducing the communication burden and maintaining excellent system performance. Therefore, we design an adaptive event-triggered method that adaptively adjusts the trigger threshold according to the consistency error. Finally, a simulation system is established. The proposed control strategy has been validated through simulation results.

## 1. Introduction

### 1.1. Motivation and incitement

MICROGRIDS include distributed generators, energy conversion equipment, loads, surveillance, and protection equipment [1–3]. MGs usually use a hierarchical control scheme, generally divided into three layers [4,5]. In this article, we mainly pay attention to the secondary control.

The distributed control method has attracted the attention of more and more scholars [6,7]. It has been widely used in voltage and frequency restore control in MGs [8]. However, the limited bandwidth and the communication delay are inevitable. The common problems in these two network controls will affect the performance of distributed secondary controllers and even cause the system to collapse.

### 1.2. Literature review

In [9], the influence of delay on the secondary control of MGs was studied. A distributed stochastic secondary controller is proposed to ensure a distributed secondary control performance under communication delay [10]. In [11], the robust stability of secondary voltage control of the MGs under fixed communication delay is considered. In [12], the secondary control based on event-triggered for MGs systems with a single delay was studied. A robust control method is provided for the time-varying delay in communication networks for AC MGs [13,14]. In [15], a novel weighted average prediction (WAP) controller is offered to compensate for the delayed system state. However, in the above literature [9–15], the communication delay between DGs is simplified into a single time delay system, that is, $\tau = \tau_1 = \cdots = \tau_n$. Due to the decentralized property of DGs in MGs, wireless communication networks are widely used in MGs for information transmission, which inevitably causes time delays [16,17]. The islanded MGs are usually built in mountainous areas or island areas with complex natural environments. The height of the terrain and the existence of mountains can hinder the transmission of wireless signals. The terrain fluctuation can also cause multipath effects in signal transmission and lead to communication delays. Due to the wide range of decentralized DGs, each DG's natural environment differs, resulting in inconsistent communication delays between DGs. The communication delays of adjacent DGs are not necessarily the same, so multiple delays in the topological communication network of MGs are established, and the stability for multiple time-delayed is investigated [18]. In [19], the robust stability of secondary voltage control of islanded MGs is studied under single delay and independent multiple time delay. In [20,21], the impact of single time delay and multiple time delays on the consistency of MGs is studied, and both the design and verification processes are carried out. Compared with simple time delays, complex multi time delay systems are more closely related to the actual operating environment of microgrids.

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**Nomenclature**

| | |
|---|---|
| $\omega_i, V_i$ | Output voltage and frequency of DGi. |
| $V_{ni}, \omega_{ni}$ | Set values of frequency and voltage of DGi. |
| $m_P, n_Q$ | Droop coefficients of frequency and voltage. |
| $P_i, Q_i$ | Active power and reactive power of DGi. |
| $u_{\omega i}, u_{vi}$ | Frequency and voltage control inputs of DGi. |
| $Z_{\omega i}, Z_{vi}$ | Control outputs of DGi. |
| $\Phi_{\omega i}, \Phi_{vi}$ | Frequency and voltage event-triggered matrix of DGi. |
| $\varepsilon_{\omega i}, \varepsilon_{vi}$ | Frequency and voltage measurement error of DGi. |

However, the multiple time delay issue is much more complicated. The problem of limited communication bandwidth has not been addressed in the abovementioned research on multi-time delay systems.

The event-triggered scheme is used in distributed secondary control to solve low-bandwidth communication problems. In [22], event-triggered-based secondary control was proposed for the first time, and both voltage and frequency could be restored to reference values while the communication burden was significantly reduced. However, compared with traditional controls, event-triggered schemes exhibit more oscillations [22]. A distributed finite time event-triggered controller is proposed for voltage and frequency restore control [23, 24]. In [25], a frequency secondary controller based on event-triggered is proposed for MGs using virtual synchronous generators for primary control. This scheme considers the heterogeneous characteristics caused by different DG parameters in MGs. In DC MGs, the current sharing and voltage regulation controllers also adopt the event-triggered scheme to reduce the communication burden [26,27]. The event-triggered condition adopted in [22–27] is usually $\varepsilon^T\varepsilon \geq \delta e^T e$, and $e^T e$ approaches 0 when approaching consensus, which may lead to some unnecessary messages being sent. Recently, a dynamic event-triggered scheme has been proposed to solve this problem [28]. Dynamic event-triggered adds a dynamically changeable $\eta(t)$ ($\eta(t) > 0$) to the right side of the triggering condition inequality based on fixed threshold event-triggered. The key of dynamic event-triggered to reduce the communication burden after system stabilization is the introduction of $\eta(t)$ ($e^T e \geq \delta\varepsilon^T\varepsilon + \eta(t), \eta(t) > 0$). Current sharing and voltage recovery based on dynamic event-triggered are proposed for the first time and proved to be stable in DC MGs [29]. In [30], a distributed secondary control scheme based on dynamic event-triggered protocol for AC microgrids is proposed. In [22–30], time delay is not considered in the design of event-triggered secondary control. To ensure the effective and safe operation of the MGs system, we need to consider both the limited communication bandwidth and time delay issues. In [31], a secondary controller based on event-triggered under delay is proposed. In [32], the design of an event-triggered secondary controller under delay and external disturbance is further considered, and the stability is proved by using the Lyapunov-Krasovskii theory. In [33,34], A finite time event-triggered secondary controller is proposed. In [35], the self-triggered secondary control of MGs systems with a single delay is studied. The proposed self-triggered scheme can reduce the computation but increase the communication burden. In [36], MG distributed secondary voltage control under time-varying delay is studied, and the communication burden is reduced through event-triggered. In [37,38], the design problem of secondary controllers for islanded MGs under a hybrid system framework considering time-varying delays was studied, while the communication burden was reduced through event-triggered.

### 1.3. Contributions and paper organization

In the above research, the event-triggered scheme achieves the purpose of prefetch, but it can still improve efficiency further. On the one hand, the research in [31–38] is the design problem of event-triggered controllers with a single delay. The design of event-triggered secondary controllers of MGs with multiple delays has not yet been studied, and it will bring more complex theoretical proofs. On the other hand, the threshold of ETC is constant in [22–27]. Traditional event-triggered with a constant threshold ($\delta$) can effectively reduce communication between distributed power sources. However, when the system undergoes drastic changes, dense communication is required between distributed power sources to ensure system performance. Event-triggered mechanisms with large thresholds become obstacles to dense communication, thereby affecting system performance. On the contrary, when the system state is stable, only a small amount of communication is required, and an event-triggered mechanism with small thresholds will bring a large amount of useless communication. Due to the fixed threshold $\delta$ and $\eta_i(t) > 0$, dynamic event-triggered in [28–30] can only reduce the communication frequency based on traditional event-triggered, and cannot achieve an increase in communication frequency for improving system performance. Therefore, when the system changes sharply, the dynamic event-triggered mechanism has a greater hindrance to communication than the fixed threshold event-triggered, resulting in worse dynamic performance of the system than the fixed threshold event-triggered. Therefore, the current event-triggering schemes fail to solve the problem mentioned in [20]: the system will increase more oscillations after adding event-triggering schemes. The current event-triggered scheme is inefficient because it does not achieve good communication reduction and system performance. It is necessary to propose an adaptive event-triggered secondary control strategy under multiple delays, which can adjust the threshold according to the state change, which is also the motivation of this paper.

Based on the above motivation, we propose an adaptive event-triggered $H_\infty$ distributed secondary control for AC MGs, which considers communication burdens, multiple time delays, and external disturbance. The paper's major contributions are as follows:

1) The event-triggered $H_\infty$ consensus scheme for secondary control of MGs with multiple delays is studied. The frequency and output voltages of the MGs are restored to reference values. Compared to the secondary control with multiple delays in [16–21], the communication burden is significantly reduced by using event-triggered
2) An adaptive event-triggered mechanism is proposed for secondary control of MGs. Compared with the traditional event-triggered in [22–27] and dynamic event-triggered in [28–30], the adaptive adjustment of thresholds in the communication mechanism improves the performance of the control system and further reduces the communication burden.

The rest of this paper is organized as follows. In Section II, Primary droop control and control objectives are described. The controller is designed in Section III. In Section IV, the simulation builds and compares the previous research to prove the contribution of the paper. The conclusion is in Section V.

## 2. Secondary control framework and problem statement

### 2.1. Graph theory

Let $G = (V, \varepsilon, A)$ be a directed graph containing a node set $v = \{1, 2, \cdots, N\}$, a set of edges $\varepsilon = V \times V$, and adjacency matrix $A = [a_{ij}] \in R^{nn}$. We use $(j, i) \in \varepsilon$ to show agent $i$ can obtain information from $j$. And when $(j, i) \in \varepsilon$, we define $a_{ij} = 1$. In addition, we define $a_i = 0$. The Laplacian matrix $L = [l_{ij}] \in R^{nn}$ is defined as $l_{ii} = \sum_{j=1, j\neq i}^{n} a_{ij}$ and $l_{ij} = -a_{ij}, i \neq j$. Define matrix $B = diag\{b_1, b_2, \cdots, b_N\}$, where $b_i \geq 0$ indicates whether there is communication between agent $i$ and the leader. $b_i$ satisfies the condition: if $(s, i)$ is an edge of $G, b_1 = 1$; otherwise, $b_1 = 0$.

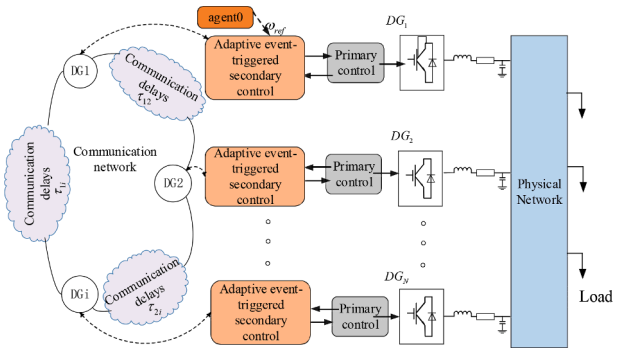



**Fig. 1.** Distributed control scheme of islanded MGs.

### 2.2. Control objectives

The droop control adjusts the frequency $\omega_i$ and the voltage $V_i$ respectively according to the active and reactive power, which can be expressed as:

$$\begin{cases} \omega_i = \omega_{ni} - m_P P_i \\ V_i = V_{ni} - n_Q Q_i \end{cases} \tag{1}$$

where $V_{ni}$ and $\omega_{ni}$ are designed in the secondary control layer. $m_P$, $n_Q$ is the frequency and voltage droop coefficient, respectively.

Fig. 1 depicts the overall structure of distributed control for islanded MGs. In this structure, communication between DGs is considered multiple delays rather than a single delay. The adaptive event-triggered method with adaptive adjustment of trigger threshold replaces the traditional fixed threshold event-triggered. Agent 0 can be considered as a virtual leader, with all other agents synchronized with it. Through this approach, the following control objectives are achieved:

1) The frequency amplitude of DGs is restored to its reference frequency. That is,

$$\lim_{t\to\infty} \omega_i(t) = \omega_{ref} \tag{2}$$

where $\omega_i$ is the frequency of $DG_i$. $\omega_{ref}$ is the reference frequency value.

2) The voltage amplitude of all DGs is restored to its reference voltage. That is,

$$\lim_{t\to\infty} V_i(t) = V_{ref} \tag{3}$$

where $V_i$ is the voltage of $DG_i$. $V_{ref}$ is the reference voltage value.

**Lemma 1.** *([10] Schur Complement): For given symmetric matrix$F_{11}$, $F_{12}$, $F_{22}$, and $F_{22} < 0$, then,*

$$\begin{bmatrix} F_{11} & F_{12} \\ F_{21} & F_{22} \end{bmatrix} < 0 \Leftrightarrow F_{11} - F_{12}F_{22}^{-1}F_{12}^{T} < 0 \tag{4}$$

## 3. Main results

### 3.1. Adaptive event-triggered $H_\infty$ secondary frequency controller

We can ignore the fast dynamics in designing the secondary control [39]. The frequency secondary control state-space model is built as

$$\begin{cases} \dot{\omega}_i(t) = u_{\omega i}(t) \\ Z_{\omega i}(t) = \omega_i(t) \end{cases} \tag{5}$$

The event-triggered distributed controllers are structured as follows:

$$u_{\omega i}(t) = K_{\omega i} e_{\omega i}(t) + d_{\omega i}(t) \tag{6}$$

$$\begin{aligned} e_{\omega i}(t) = \sum_{j\in N_i} a_{ij}\left[\omega_j\left(t_k^i - \tau_{i,j}(t)\right) - \omega_i\left(t_k^i - \tau_{i,j}(t)\right)\right] \\ + b_i\left[\omega_{ref} - \omega_i\left(t_k^i - \tau_{i,j}(t)\right)\right] \end{aligned} \tag{7}$$

where $K_{\omega i} > 0$ is the feedback control gain. $\| d_{\omega i}(t)\| \le \Phi$ is the external bounded disturbance. $e_{\omega i}(t)$ is the consistency error.

Different from the controller proposed in [31–38], the information received by DGi from different neighbour DGj at each time in the proposed controller has different delay $\tau_{i,j}(t)$. When $\tau_{i,j}(t) = \tau(t)$, the controller proposed in reference [31–38] is a special form of the proposed controller.

$\tau_{i,j}(t)$ means that agent $i$ ($i = 1, \ 2, \ \cdots \ N$) receives different delay information from different neighbour agent $j$ at $t$. Define $\tau_m(t) \in \tau_{i,j}(t)$, $m = 1, \ 2, \ \cdots \ r$, where $r > 0$ indicates the total number of different delays be $r$, $r \le n(n-1)/2$. Since $N$ DGs contain $r$ delays, the system state is decomposed into r states with reference to [16], that is, $\omega(t_k^i - \tau_1(t))$, $\omega(t_k^i - \tau_2(t))$, …, $\omega(t_k^i - \tau_r(t))$. The Laplacian matrix $L$ is divided into r Laplacian matrix $L_m$. Similar to the Laplacian matrix $L$, the matrix $L_m = [l_{mij}]$ is defined as:

$$l_{mij} = \begin{cases} -a_{ij}, & j \ne i, \tau_m(\bullet) = \tau_{ij}(\bullet) - a_{ij}, \\ -\sum_{m=1}^{n} l_{mij}, & j = i \\ 0, & j = i, \tau_m(\bullet) \ne \tau_{ij}(\bullet) \end{cases} \tag{8}$$

Rewrite the (7) as follows

$$\begin{aligned} e_{\omega i}(t) = \Bigg[ \sum_{m=1}^{r} \sum_{j\in N_i} a_{ij}\left[\omega_j\left(t_k^i - \tau_m(t)\right) - \omega_i\left(t_k^i - \tau_m(t)\right)\right] \\ + b_i\left[\omega_{ref} - \omega_i\left(t_k^i - \tau_m(t)\right)\right] \Bigg] \end{aligned} \tag{9}$$

where the delay in the distributed control system of MGs satisfies: $0 \le \tau_m(t) \le h_m, \ \ \dot{\tau}_m(t) \le d_i, \ \ h_m, \ \ d_i \ge 0$ .

Remake 1:The frequency reference value $\omega_{ref}$ (the virtual leader) is predefined for DGs, and one of the DGs is known in advance [40]. This means that the virtual leader is the local information of the DGi, and there is no communication delay between them. For the convenience of subsequent analysis, we consider $\omega_{ref} - \omega_i(t_k^i)$ as $\omega_{ref} - \omega_i(t_k^i - \tau_{i,j}(t))$.

Defining the state measurement error $\varepsilon_{\omega i}(t)$ and the global error $\eta_{\omega i}(t)$ for each $DG_i$, respectively, as:

$$\varepsilon_{\omega i}(t) = \omega_i\left(t_k^i\right) - \omega_i(t) \tag{10}$$

$$\eta_{\omega i}(t) = \omega_i(t) - \omega_{ref} \tag{11}$$

Combining the above definitions, the system model of the frequency secondary control is structured as

$$\begin{cases} \dot{\eta}_\omega(t) = -K\sum_{m=1}^{r}\left[(L_m + B)(\varepsilon_\omega(t-\tau_m(t)) + \eta_\omega(t-\tau_m(t)))\right] \\ \qquad\qquad + D_\omega(t) \\ Z_\omega(t) = U\eta_\omega(t) \end{cases} \tag{12}$$

where $\eta_\omega = [\eta_{\omega 1}, \ \eta_{\omega 2}, \ \cdots, \ \eta_{\omega N}]^T$, $\varepsilon_\omega = [\varepsilon_{\omega 1}, \ \varepsilon_{\omega 2}, \ \cdots, \ \varepsilon_{\omega N}]^T$, $D_\omega = [d_{\omega 1}, \ d_{\omega 2}, \ \cdots, \ d_{\omega N}]^T$, $Z_\omega = [z_{\omega 1}, \ z_{\omega 2}, \ \cdots, \ z_{\omega N}]^T$, $K = diag(k_\omega . I_n)$.

It should be noted that the controller in (6) (8) is only modified at the trigger instant $t_k^i$. The event-triggered conditions are structured as

$$t_k^i = \inf\{t > t_{k-1}^i | f_i(t) \geq 0\}, t \in [t_k^i, \quad t_{k+1}^i] \tag{13}$$

where the trigger function of DGi is

$$f_i(t) = \sum_{m=1}^{r} \varepsilon_{\omega i}^T(t-\tau_m(t))\Phi_{\omega i}\varepsilon_{\omega i}(t-\tau_m(t)) - \delta_{\omega i}e_{\omega i}(t)^T\Phi_{\omega i}e_{\omega i}(t) \tag{14}$$

To consider the multi time delays, the trigger function is the sum of measurement errors under different delays. When $m = 1$, the trigger function (14) degenerates to the traditional single time delay trigger function, that is, $f_i(t) = \varepsilon_{\omega i}^T(t-\tau_1(t))\Phi_{\omega i}\varepsilon_{\omega i}(t-\tau_1(t)) - \delta_{\omega i}e_{\omega i}(t)^T\Phi_{\omega i}e_{\omega i}(t)$

Remake 2:In (13) (14), we design an event-triggered trigger function for multiple time delays. But like traditional event-triggered in [20–29], it has a fixed threshold $\delta_{\omega i}$. The threshold affects the event-triggered performance (communication burden) and the system's performance. When the threshold is small, the maximum delay that can be tolerated increases and the robust stability of the system improves, but the communication burden increases. When the threshold is large, the communication times decrease and the event-triggered performance improves, but the robust stability of the system deteriorates. Therefore, the traditional event-triggered with fixed-parameter cannot achieve the optimal event-triggered performance and robust stability of the system.

From the above analysis, we need to lower the threshold when the system is unstable and raise the threshold when the system is stable. Especially for islanded MGs, load fluctuation and plug-and-play problems often exist in its operation, which will lead to system instability.

Therefore, we propose an adaptive event-triggered method that adaptively adjusts the threshold according to the convergence or divergence trend of the consistency error, and the adaptive event-triggered conditions are redesigned as

$$t_k^i = \inf\{t > t_{k-1}^i | f_i(t) \geq 0, |e_{\omega i}(t)| \geq s_\omega\}, t \in [t_k^i, \quad t_{k+1}^i] \tag{15}$$

where the trigger function of DGi is

$$f_i(t) = \sum_{m=1}^{r} \varepsilon_{\omega i}^T(t-\tau_m)\Phi_{\omega i}\varepsilon_{\omega i}(t-\tau_m) - \delta_{\omega i}(t)e_{\omega i}(t)^T\Phi_{\omega i}e_{\omega i}(t) \tag{16}$$

Note that the trigger threshold in the above equation is time-varying rather than the traditional fixed threshold, i.e., $\delta_{\omega i}$ was replaced by $\delta_{\omega i}(t)$. Then, $\delta_{\omega i}(t)$ is determined by the following adaptive adjustment mechanism:

$$\begin{aligned} &\delta_{\omega i}(t) = \min\{\delta_{\omega\max} \quad , \quad \lambda_i\delta_{\omega i}(t)\}, \quad 0 < \delta_{\omega\max} < 1 \\ &\lambda_i = 1 - \frac{2\alpha_i}{\pi}\arctan\left\{\frac{\| e_{\omega\max i}^k(t) \| - \| e_{\omega\max i}^{k-1}(t) \|}{\| e_{\omega\max i}^{k-1}(t) \|}\right\} \end{aligned} \tag{17}$$

where $\delta_{\omega\max}$ is the maximum upper bound of the threshold. $\alpha_i > 0$ can change the rate of change of the threshold. $e_{\omega\max i}^k(t)$ is the th $k$ local extreme points of the frequency consistency error of the $DG_i$. The threshold adjustment parameter $\lambda_i$ represents the incremental direction of two adjacent local extreme points, which can mean the convergence or divergence trend of the system. For example, if $\| e_{\omega\max i}^k(t)\| < \| e_{\omega\max i}^{k-1}(t)\|$ indicates that the system tends to diverge, then the trigger threshold of $\lambda_i \in (1, \mathbf{1}+\alpha_i)$ decreases, and the trigger frequency increases accordingly, and vice versa. This allows the system to maintain stability with the least amount of communication. Adjust $\alpha_i$ to change the rate of change of the threshold.

Remake 3: Compared to traditional event-triggered functions, the condition of $|e_{\omega i}(t)| \geq s_\omega$ is added in (15) to eliminate frequent triggering near steady state, and $s_\omega$ is a small positive number.

**Theorem 1**. *Under event-triggered method (15) (16) (17), secondary frequency controllers (6) with the feedback gain matrix $K = YX^{-1}$, if there are symmetric positive matrices $X > 0$, $\widetilde{Q}_m > 0$, $\widetilde{R}_m > 0 (m = 1, 2\cdots r)$, $\widetilde{\Phi}_\omega > 0$ and $\widetilde{\Phi}_{\omega\delta} > 0$, such that*

$$\begin{bmatrix} \Xi_{11} & * \\ \Xi_{21} & \Xi_{22} \end{bmatrix} < 0 \tag{18}$$

hold, then the system (12) consensus can be reached with a prescribed $H_\infty$ performance level $\gamma > 0$, where,

$$\Xi_{21} = [\, UX \quad 0 \quad \cdots \quad 0 \quad 0 \quad \cdots \quad 0 \quad 0\,],$$

$$\Xi_{22} = -I_N,$$

$$\Xi_{11} = \begin{bmatrix} \widetilde{\Sigma}_{11} & * & * & * & * \\ \widetilde{\Sigma}_{21} & \widetilde{\Sigma}_{22} & * & * & * \\ \widetilde{\Sigma}_{31} & \widetilde{\Sigma}_{32} & \widetilde{\Sigma}_{33} & * & * \\ I_N & 0 & 0 & -\gamma^2 & * \\ 0 & \widetilde{\Sigma}_{52} & \widetilde{\Sigma}_{53} & \sum_{m=1}^{r} h_m & \widetilde{\Sigma}_{55} \end{bmatrix}$$

$$\widetilde{\Sigma}_{11} = -\sum_{m=1}^{r}\widetilde{Q}_m + \sum_{m=1}^{r}\widetilde{R}_m,\ \widetilde{\Sigma}_{21} = -\sum_{m=1}^{r}(L_m+B)^T \otimes Y^T + \sum_{m=1}^{r}\widetilde{Q}_m,$$

$$\widetilde{\Sigma}_{22} = -\sum_{m=1}^{r}\widetilde{Q}_m + \sum_{m=1}^{r}(L_m+B)^T\widetilde{\Phi}_\delta(L_m+B) - \sum_{m=1}^{r}(1-d_m)\widetilde{R}_m,$$

$$\widetilde{\Sigma}_{31} = -\sum_{m=1}^{r}(L_m+B)^T \otimes Y^T, \quad \widetilde{\Sigma}_{32} = \sum_{m=1}^{r}(L_m+B)^T\widetilde{\Phi}_\delta(L_m+B),$$

$$\widetilde{\Sigma}_{33} = \sum_{m=1}^{r}(L_m+B)^T\widetilde{\Phi}_\delta(L_m+B) + \widetilde{\Phi}, \quad \widetilde{\Sigma}_{52} = -\sum_{m=1}^{r}h_m(L_m+B) \otimes Y,$$

$$\widetilde{\Sigma}_{53} = -\sum_{m=1}^{r}h_m(L_m+B) \otimes Y, \quad \widetilde{\Sigma}_{55} = -X\widetilde{Q}_m^{-1}X,$$

***Proof***

In the following, we omit the subscript $\omega$ for convenience. We define the Lyapunov–Krasovskii function is chosen as follows:

$$\begin{aligned} V(t) = \eta^T(t)P\eta(t) &+ \sum_{m=1}^{r}\int_{-\tau_m(t)}^{0}\int_{t+s}^{t}\dot{\eta}^T(\theta)Q_m\eta(\theta)d\theta ds \\ &+ \sum_{m=1}^{r}\int_{t-\tau_m(t)}^{t}\eta^T(\theta)R_m\eta(\theta)d\theta \end{aligned} \tag{19}$$

Then, the time derivative of (19), and add and subtract $\sum_{m=1}^{r}\varepsilon_{\omega i}^T(t-\tau_m)V_{\omega i}\varepsilon_{\omega i}(t-\tau_m)$ at the same time, we can yield:

$$\begin{aligned} &\dot{V}(t) = 2\eta^T(t)P\dot{\eta}(t) + \sum_{m=1}^{r}\tau_m(t)\dot{\eta}^T(t)Q_m\dot{\eta}(t) \\ &-\sum_{m=1}^{r}\int_{t-\tau_m(t)}^{t}\dot{\eta}^T(s)Q_m\dot{\eta}(s)ds + \sum_{m=1}^{r}\eta^T(t)R_m\eta(t) \\ &-\sum_{m=1}^{r}(1-\dot{\tau}_m(t))\eta^T(t-\tau_m(t))R_m\eta(t-\tau_m(t)) \\ &+Z^T(t)Z(t) - Z^T(t)Z(t) \\ &+\sum_{m=1}^{r}\varepsilon_{\omega i}^T(t-\tau_m)\Phi_{\omega i}\varepsilon_{\omega i}(t-\tau_m) \\ &-\sum_{m=1}^{r}\varepsilon_{\omega i}^T(t-\tau_m)\Phi_{\omega i}\varepsilon_{\omega i}(t-\tau_m) \end{aligned} \tag{20}$$

Combining formula (12) yields

$$\begin{aligned}\dot{V}(t) &= \eta^T(t)P\dot{\eta}(t) + \sum_{m=1}^{r}\tau_m(t)\dot{\eta}^T(t)Q_m\dot{\eta}(t)\\ &-\sum_{m=1}^{r}h_i^{-1}[\varepsilon_\omega(t)-\varepsilon_\omega(t-\tau_m(t))]^TQ_m[\varepsilon_\omega(t)-\varepsilon_\omega(t-\tau_m(t))]\\ &+\sum_{m=1}^{r}\eta^T(t)R_m\eta(t)\\ &-\sum_{m=1}^{r}(1-d_i)\eta^T(t-\tau_m(t))R_m\eta(t-\tau_m(t))\\ &+Z^T(t)Z(t)-Z^T(t)Z(t)+\varepsilon^T(t)\Phi\varepsilon(t)-\varepsilon^T(t)\Phi\varepsilon(t)\end{aligned} \tag{21}$$

Combing with event-triggered (16) yields

$$\begin{aligned}&\sum_{m=1}^{r}\varepsilon_\omega(t-\tau_m)\Phi\varepsilon_\omega(t-\tau_m)\\ &<\sum_{m=1}^{r}[(L_m+B)(\varepsilon_\omega(t-\tau_m)+\eta_\omega(t-\tau_m))]^T\Phi_\delta\\ &\sum_{m=1}^{r}[(L_m+B)(\varepsilon_\omega(t-\tau_m))+\eta_\omega(t-\tau_m))]\end{aligned} \tag{22}$$

where, $\Phi_\delta = diag\{\delta_{\omega1}\Phi_{\omega1},\ \delta_{\omega2}\Phi_{\omega2},\ \cdots,\ \delta_{\omega N}\Phi_{\omega N}\}$, $\Phi = diag\{\Phi_{\omega1},\ \Phi_{\omega2},\ \cdots,\ \Phi_{\omega N}\}$.

Combining (12), (21), and (22), we can yield.

$$\dot{V}(t) \le \xi^T(t)\left(\Sigma+\Lambda^TQ_m\Lambda\right)\xi(t) - Z^T(t)Z(t) + \gamma^2D^T(t)D(t) \tag{23}$$

where,

$$\xi_\omega^T = \left[\eta_\omega^T(t)\quad \sum_{m=1}^{r}\eta_\omega^T(t-\tau_m(t))\quad \sum_{m=1}^{r}\varepsilon_\omega^T(t-\tau_m(t))\quad D_\omega^T(t)\right]$$

$$\Lambda = \left[0\quad -K\sum_{m=1}^{r}h_m(L_m+B)\quad -K\sum_{m=1}^{r}h_m(L_m+B)\quad \sum_{m=1}^{r}h_m\right]$$

$$\Sigma = \begin{bmatrix}\Sigma_{11} & * & * & *\\ \Sigma_{21} & \Sigma_{22} & * & *\\ \Sigma_{31} & \Sigma_{32} & \Sigma_{33} & *\\ 2P & 0 & 0 & \Sigma_{44}\end{bmatrix},$$

$$\Sigma_{11} = -\sum_{m=1}^{r}Q_m + \sum_{m=1}^{r}R_m + U^TU,$$

$$\Sigma_{21} = -\sum_{m=1}^{r}(L_m+B)^T\otimes\left(K^TP\right) + \sum_{m=1}^{r}Q_m,$$

$$\Sigma_{22} = -\sum_{m=1}^{r}Q_m + \sum_{m=1}^{r}(L_m+B)^T\Phi_\delta(L_m+B) - \sum_{m=1}^{r}(1-d_m)R_m,$$

$$\Sigma_{31} = -\sum_{m=1}^{r}(L_m+B)^T\otimes\left(K^TP\right),\ \Sigma_{32} = \sum_{m=1}^{r}(L_m+B)^T\Phi_\delta(L_m+B)$$

$$\Sigma_{33} = \sum_{m=1}^{r}(L_m+B)^T\Phi_\delta(L_m+B),\ \Sigma_{44} = -\gamma^2$$

According to the Schur complement lemma, the Lyapunov-Krasovskii function in Eq. (19) guarantees $\dot{V}(t) \le 0$ in Eq. (20). At the same time, we can easily prove that the closed-loop system (13) is asymptotically stable under $d_i(t) \equiv 0$ and $\| Z(t) \| \le \gamma \| D(t) \|$ holds under zero initial conditions.

Let $\Pi = \Sigma + \Lambda^TQ_m\Lambda$. Define $X = P^{-1}$ .Let $\widehat{X} = diag\{X, X, X, X, I, I\}$. Noting that $\widetilde{P} = XPX$, $\widetilde{Q}_m = XQ_mX$, $\widetilde{R}_m = XR_mX$ $(m = 1, 2\cdots r)$ and $Y = KX$, according to the properties of positive-definite matrix, and combined with Lemma 1, we can get:

$$\begin{aligned}&\dot{V} < 0\\ &\Leftrightarrow \Pi < 0\\ &\Leftrightarrow \begin{bmatrix}\Sigma & *\\ \Lambda & -Q_m^{-1}\end{bmatrix} < 0\\ &\Leftrightarrow \widetilde{\Pi} = \widehat{X}^T\Pi\widehat{X}\\ &= \begin{bmatrix}\widetilde{\Sigma}_{11}+X^TU^TUX & * & * & * & *\\ \widetilde{\Sigma}_{21} & \widetilde{\Sigma}_{22} & * & * & *\\ \widetilde{\Sigma}_{31} & \widetilde{\Sigma}_{32} & \widetilde{\Sigma}_{33} & * & *\\ I_N & 0 & 0 & -\gamma^2 & *\\ 0 & \widetilde{\Sigma}_{52} & \widetilde{\Sigma}_{54} & \sum_{m=1}^{r}h_m & \widetilde{\Sigma}_{55}\end{bmatrix}\end{aligned} \tag{24}$$

Thus, we obtain the design method of controller gain.

### 3.2. *Adaptive event-triggered $H_\infty$ secondary voltage controller*

The voltage secondary control state-space model is constructed as

$$\begin{cases}\dot{V}_i(t) = u_{vi}(t)\\ Z_{vi}(t) = V_i(t)\end{cases} \tag{25}$$

The adaptive event-triggered $H_\infty$ secondary voltage controller $u_{vi}$ is designed under multiple time delays as follows:

$$u_{vi}(t) = K_ve_{vi}(t) + d_i(t) \tag{26}$$

where $K_v > 0$ is the control gain. $\| d_i(t)\| \le \Phi$ is the external disturbance. $e_{vi}(t)$ is defined as:

$$\begin{aligned}e_{vi}(t) = &\left[\sum_{m=1}^{r}\sum_{j\in N_i}a_{ij}\left[V_j\left(t_k^i-\tau_m(t)\right) - V_i\left(t_k^i-\tau_m(t)\right)\right]\right.\\ &\left.+b_i\left[V_{ref} - V_i\left(t_k^i-\tau_m(t)\right)\right]\right]\end{aligned} \tag{27}$$

where $\tau_m(t) \in \tau_{i,j}(t), m = 1,\ 2,\ \cdots\ r$ means that there are $r$ communication delays at $t$ time. The total number of different delays be $r$, $r \le n(n-1)/2$. And it is assumed that the delay in the distributed control system of MGs (26) (27) satisfies: $0 \le \tau_m(t) \le h_m,\ \dot{\tau}_m(t) \le d_i,\ h_m,\ d_i \ge 0$.

Then, the adaptive event-triggered method is designed for secondary voltage control as

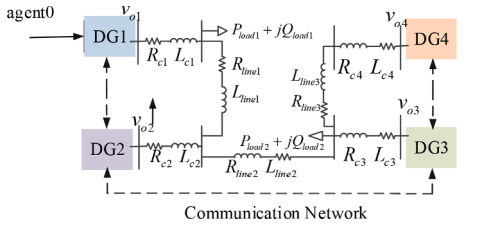



**Fig. 2.** Electrical and communication connections for MGs simulation systems.

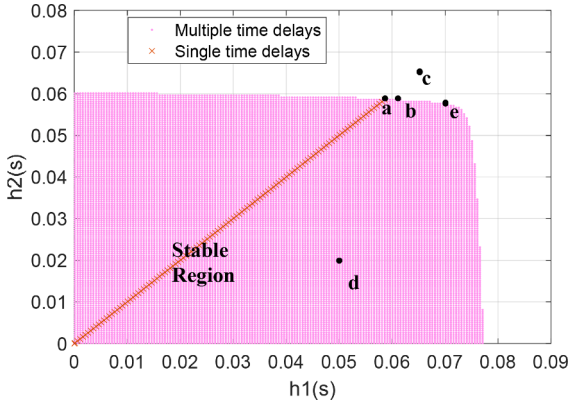



**Fig. 3.** The stability region of the secondary control.

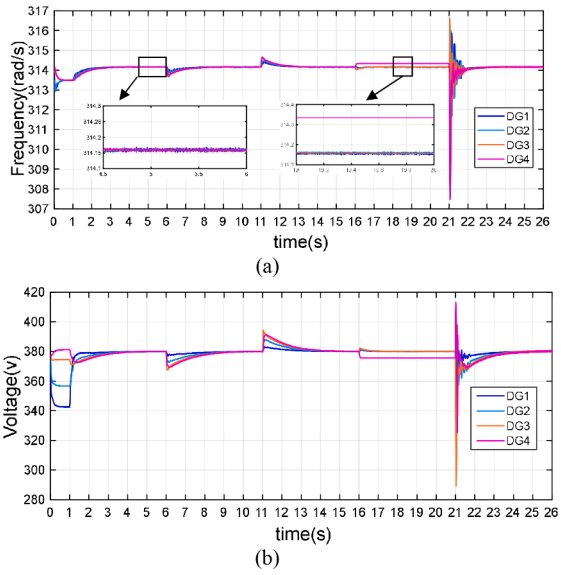



**Fig. 4.** Performance of event-triggered secondary controllers. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.).

$$t_k^i = \inf\{t > t_{k-1}^i | f_i(t) \geq 0, |e_{vi}(t)| \geq s_v\}, t \in [t_k^i, \quad t_{k+1}^i] \tag{28}$$

where the trigger function of DGi is

$$f_i(t) = \sum_{m=1}^{r} \varepsilon_{vi}^T(t-\tau_m)\Phi_{vi}\varepsilon_{vi}(t-\tau_m) - \delta_{vi}(t)e_{vi}(t)^T\Phi_{vi}e_{vi}(t) \tag{29}$$

and, where $\varepsilon_{vi}(t)$ is defined as

$$\varepsilon_{vi}(t) = V_i(t_k^i) - V_i(t) \tag{30}$$

Then, $\delta_{vi}(t)$ is determined by the following adaptive adjustment mechanism:

$$\begin{aligned} &\delta_{vi}(t) = \min\{\delta_{v\max} \quad , \quad \lambda_i\delta_{vi}(t)\}, \quad 0 < \delta_{v\max} < 1 \\ &\lambda_i = 1 - \frac{2\alpha_i}{\pi}\arctan\left\{\frac{\| e_{v\max i}^k(t) \| - \| e_{v\max i}^{k-1}(t) \|}{\| e_{v\max i}^{k-1}(t) \|}\right\} \end{aligned} \tag{31}$$

where $\delta_{v\max}$ is the maximum upper bound of the threshold. $e_{v\max i}^k(t)$ is the th $k$ local extreme points of the voltage consistency error of the DGi.

In combination with the above formula, we define the global error, $\eta_{vi}(t) = V_i(t) - V_{ref}$. We can get the system model of the distributed voltage secondary control based on the adaptive event-triggered as follows

$$\begin{cases} \dot{\eta}_v(t) = -K\sum_{m=1}^{r}[(L_m + B)(\varepsilon_v(t-\tau_m) + \eta_v(t-\tau_m))] \\ \qquad\qquad +D_v(t) \\ Z_v(t) = U\eta_v(t) \end{cases} \tag{32}$$

where, $\eta_v = [\eta_{v1}, \quad \eta_{v2}, \quad \cdots, \quad \eta_{vN}]^T$, $\varepsilon_v = [\varepsilon_{v1}, \quad \varepsilon_{v2}, \quad \cdots, \quad \varepsilon_{vN}]^T$, $D_v = [d_{v1}, \quad d_{v2}, \quad \cdots, \quad d_{vN}]^T$, $Z_v = [z_{v1}, \quad z_{v2}, \quad \cdots, \quad z_{vN}]^T$, $K_v = diag(k_v.I_n)$.

**Theorem 2.** *Under adaptive event-triggered method (28) (29), secondary voltage controllers (26) (27) with the feedback gain matrix $K = YX^{-1}$, if there are symmetric positive matrices $X > 0$, $\widetilde{Q}_m > 0$, $\widetilde{R}_m > 0 (m = 1, 2\cdots r)$, $\widetilde{\Phi}_v > 0$ and $\widetilde{\Phi}_{v\delta} > 0$, such that*

$$\begin{bmatrix} \Xi_{11} & * \\ \Xi_{21} & \Xi_{22} \end{bmatrix} < 0 \tag{33}$$

hold, then the system (32) consensus can be reached with a prescribed $H_\infty$ performance level $\gamma > 0$, where

$$\xi_v^T = \begin{bmatrix} \eta_v^T(t) & \sum_{m=1}^{r}\eta_v^T(t-\tau_m(t)) & \sum_{m=1}^{r}\varepsilon_v^T(t-\tau_m(t)) & D_v^T(t) \end{bmatrix}$$

$$\Xi_{21} = [UX \quad 0 \quad \cdots \quad 0 \quad 0 \quad \cdots \quad 0 \quad 0]$$

$$\Xi_{22} = -I_N,$$

$$\Xi_{11} = \begin{bmatrix} \widetilde{\Sigma}_{11} & * & * & * & * \\ \widetilde{\Sigma}_{21} & \widetilde{\Sigma}_{22} & * & * & * \\ \widetilde{\Sigma}_{31} & \widetilde{\Sigma}_{32} & \widetilde{\Sigma}_{33} & * & * \\ I_N & 0 & 0 & -\gamma^2 & * \\ 0 & \widetilde{\Sigma}_{52} & \widetilde{\Sigma}_{53} & \sum_{m=1}^{r}h_m & \widetilde{\Sigma}_{55} \end{bmatrix}$$

$$\widetilde{\Sigma}_{11} = -\sum_{m=1}^{r}\widetilde{Q}_m + \sum_{m=1}^{r}\widetilde{R}_m, \widetilde{\Sigma}_{21} = -\sum_{m=1}^{r}(L_m + B)^T \otimes Y^T + \sum_{m=1}^{r}\widetilde{Q}_m,$$

$$\widetilde{\Sigma}_{22} = -\sum_{m=1}^{r}\widetilde{Q}_m + \sum_{m=1}^{r}(L_m + B)^T\widetilde{\Phi}_{v\delta}(L_m + B) - \sum_{m=1}^{r}(1-d_m)\widetilde{R}_m$$

$$\widetilde{\Sigma}_{31} = -\sum_{m=1}^{r}(L_m + B)^T \otimes Y^T, \widetilde{\Sigma}_{32} = \sum_{m=1}^{r}(L_m + B)^T\widetilde{\Phi}_{v\delta}(L_m + B),$$

$$\widetilde{\Sigma}_{33} = \sum_{m=1}^{r}(L_m + B)^T\widetilde{\Phi}_{v\delta}(L_m + B) + \Phi, \widetilde{\Sigma}_{52} = -\sum_{m=1}^{r}h_m(L_m + B) \otimes Y,$$

$$\widetilde{\Sigma}_{53} = -\sum_{m=1}^{r}h_m(L_m + B) \otimes Y, \widetilde{\Sigma}_{55} = -X\widetilde{Q}_m^{-1}X$$

## 4. Simulation results

To evaluate the performance of the proposed secondary controller, we constructed an islanded MGs simulation system according to Fig. 2. The parameters of the islanded MGs are as follows: $n_{Q1-2} = 3.5 \times 10^{-5}$, $n_{Q3-4} = 2.7 \times 10^{-5}$, $m_{p1-2} = 3 \times 10^{-5}$, $m_{p1-2} = 2 \times 10^{-5}$.Other parameters of the MGs can be referenced in [35].

### 4.1. Performance of the proposed secondary controller

The communication delay between DGs is assumed to be $\tau_{1j}(t) = \tau_1$, $\tau_{ij}(t) = \tau_2$, i.e., $m = 2$. In order to facilitate analysis, we set the parameters of voltage and frequency control as the same. The external disturbance for DGs is established as: $D = [0.5d \quad d \quad d \quad -0.5d]$, $d = e^{-0.47t}$. And by selecting the parameters $\gamma = 1$, $K_\omega = K_v = [10 \quad 10 \quad 10 \quad 10]$, $s_\omega = 0.024$, $s_v = 0.39$, $\delta_\omega = \delta_v = 0.059$,$d_{\omega 1} = d_{v1} = 1, d_{\omega 2} = d_{v2} = 1$ and solving the LMI, we can obtain the stability region of frequency and voltage secondary control of MGs under the multiple time delay. From Fig. 3, we can get the maximum delay value under multiple time delays$h_m (m = 1, 2)$, $h_1 \leq 0.0601$, $h_2 \leq 0.0771$.

#### 4.1.1. Performance of the proposed secondary controller

We select point d (h1=0.05 s, h2=0.02 s) in the multi-delay stable region to verify the controller performance. The following constitutes the simulation scenario:

1. At $t = 0s$, all DGs and load-1 ($P_{load1} = 70kW$, $Q_{load1} = 70kVar$) are connected to the MGs. The primary control is activated for MGs.
2. At $t = 1s$, the secondary controller is activated for MGs.
3. At $t = 6s$, The Load-2 ($P_{load1} = 70kW$, $Q_{load1} = 100kVar$) are connected to the MGs. And, the Load-2 has been disconnected in its connection with the MGs at $t = 11s$.
4. At $t = 16s$, The DG4 is disconnected from the MGs. And, the DG4 has been reestablished in its connection with the MGs at $t = 21s$.

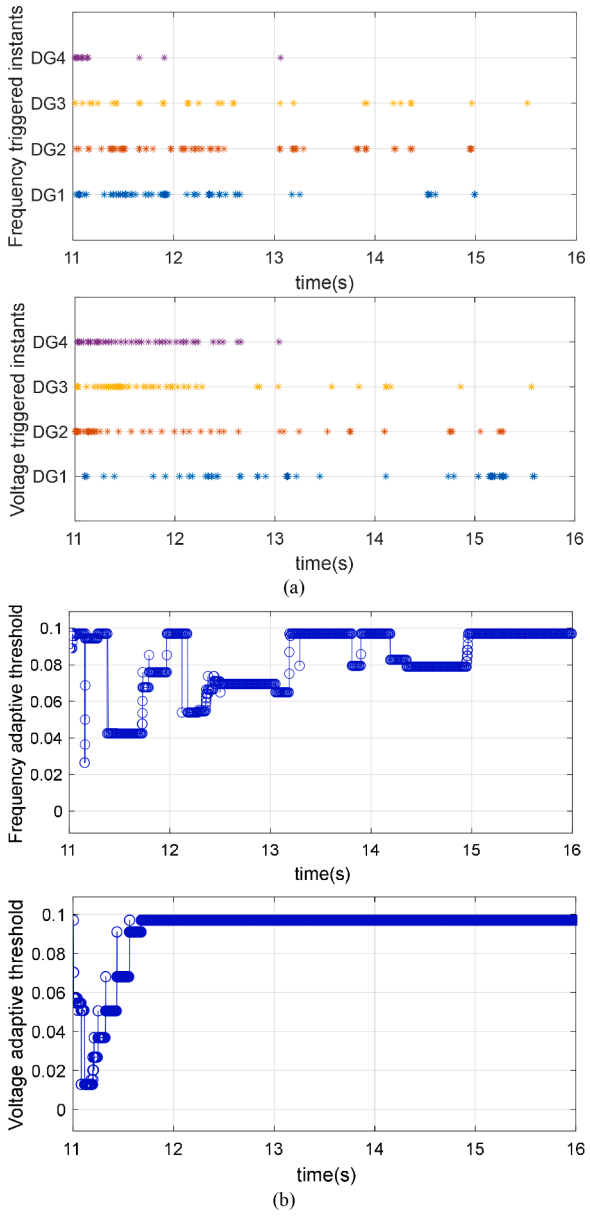



**Fig. 5.** The performance of adaptive event-triggered. a triggered instants. b Adaptive threshold for DG2). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.).

Before activating the secondary controller, both frequency and voltage deviate from the reference values, and the voltage of each distributed power source is not consensus. In Fig. 4(a), the frequency restores to the reference ($\omega_{ref} = 314.159$) after the controller is activated at time $t = 1$ s. When $t = 6$ s and $t = 11$ s, the frequency amplitude fluctuates due to the load change. However, the frequency can return to the reference frequency under the proposed controller's administration. Similarly, in Fig. 4(b), whenever the voltage amplitude fluctuates, it can be restored to the reference voltage $V_{ref} = 380$.

Through repeated disconnection and reconnection of DG4, we were able to evaluate the "plug and play" functionality of the proposed controller. In Fig. 4, the frequency and voltage can still return to the value set as a reference after the disconnection of DG4 ($t = 16$ s) and subsequent reconnection ($t = 21$ s). As a result, the suggested secondary controller can realize the control goals of recovery frequency and voltage in the event of plug-and-play.

Fig. 5 shows the performance of the adaptive event-triggered frequency secondary controller during the period from system fluctuation due to load-2 disconnection to system stabilization. Fig. 5(a) shows the event-triggered instants in the control process. As the system continues to converge after the load fluctuates, there are fewer and fewer triggered instants, and it is clear that the number of updates sent to the DG controller follows a sporadic pattern.

Correspondingly, as shown in Fig. 5(b), the threshold of adaptive event-triggered rapidly decreases when the load fluctuates. And the threshold value of adaptive event-triggered gradually increases to the maximum value when the system tends to be stable.

#### 4.1.2. *Verification and comparison of stable region*

Different time delay points were used for simulation verification of the control system: b ($h_1 = 0.0616s, h_2 = 0.0586s$), c ($h_1 = h_2 = 0.065s$), and e ($h_1 = 0.0701s, h_2 = 0.0576s$). We use the same simulation scenarios and parameters as in Section A. From the results in Fig. 6, we can see that the frequency and voltage have recovered to $\omega_{ref}$ and $V_{ref}$ when the maximum delay of the system is set to points b and e. However, when the delay was set to point c, the frequency and voltage of the MGs could not converge consistently. These results agree with the stability region depicted in Fig. 3, which proves that the proposed adaptive event-triggered secondary control strategy for multiple time delays is successful.

The delay problem considered in the event-triggered secondary control of MGs is usually a single delay in [31–38]. The stable region under a single delay is shown in Fig. 3, and the maximum delay under the same control strategy and control gain as with multiple delays is point a (point a, $h_1 = h_2 = 0.0586s$). Compared with point a, point b ($h_1 = 0.0616s, h_2 = 0.0586s$) has a larger upper limit of $h_1$ under the same delay $h_2$. Therefore, the controller designed based on multiple delays in this article is less conservative than that based on single delay systems.

Compared with point a, point e ($h_1 = 0.0701s, h_2 = 0.0576s$) can increase $h_1$ by 11.5 ms if $h_2$ drops by 1 ms. This has obvious advantages for the MGs formed by multiple DGs. The DGs in MGs are not uniformly distributed. Under the same controller parameter configuration, the proposed control scheme can tolerate larger communication delays for distributed power supply in remote and isolated areas. It can be seen that considering multiple delays can help one cope with the actual operating environment better than the single delay.

### 4.2. *Comparison with the traditional event-triggered*

The relevant results of secondary control under multiple time delays are usually based on periodic state feedback control [16–19]. However, this scheme relies on periodic communication, which is neither economical nor efficient because of its high communication burden. Therefore, the adaptive event-triggered communication mechanism is proposed to reduce the communication burden, and the threshold is adjusted according to the system's convergence and divergence to improve the control system's performance. At present, traditional event-triggered and dynamic event-triggered are commonly used to reduce the communication burden in other studies of microgrid secondary control. We consider periodic communication in [16–19], traditional event-triggered in [22–27], and dynamic event-triggered in [28–30] as contrast schemes.

We select point a and calculate the upper bound of the event-triggered threshold of this delay point as $\delta_{\omega\max} = \delta_{v\max} = 0.097$. We use the same simulation scenarios and parameters as in section A.

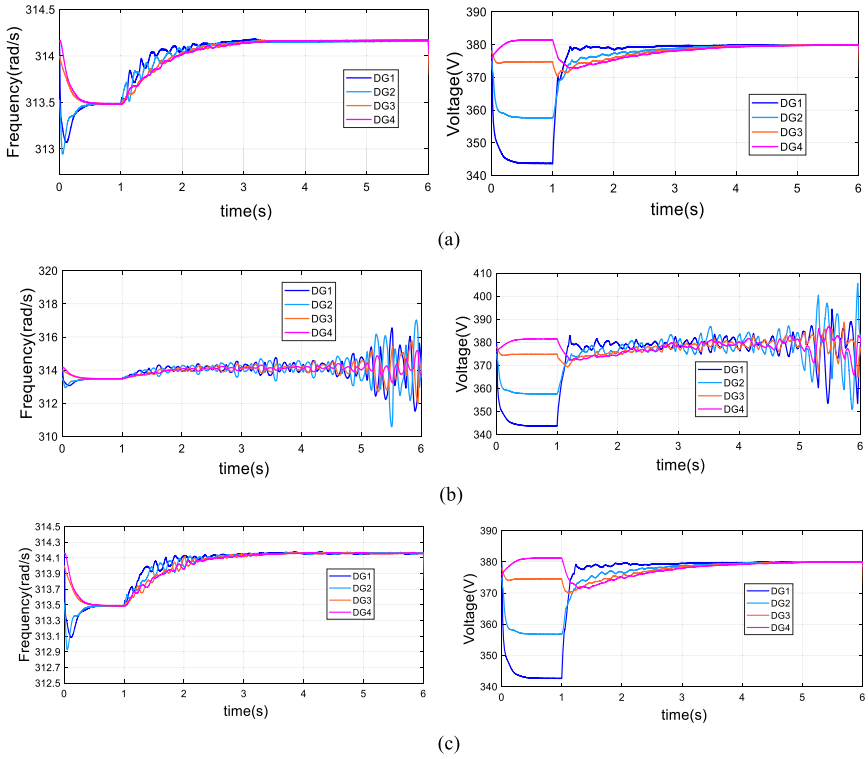



**Fig. 6.** Responses of MG with different multiple time delays. (a) point b. (b) point c. (c) point e.

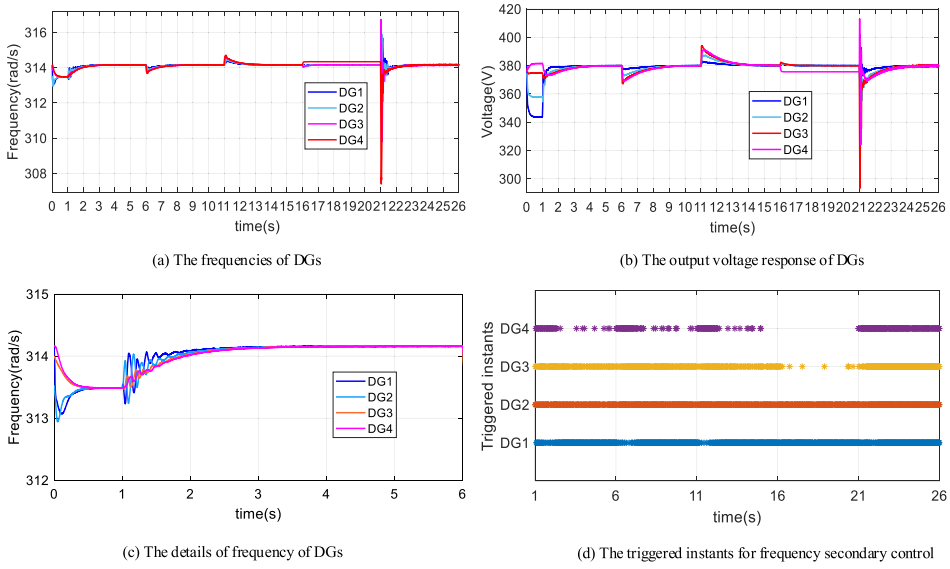
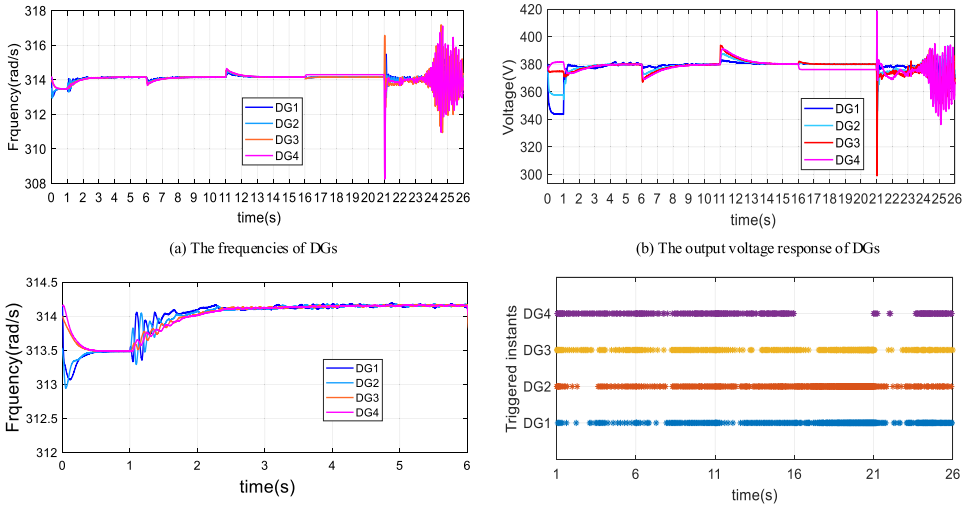



**Fig. 7.** The simulation results under the traditional event-triggered with fixed threshold. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.).



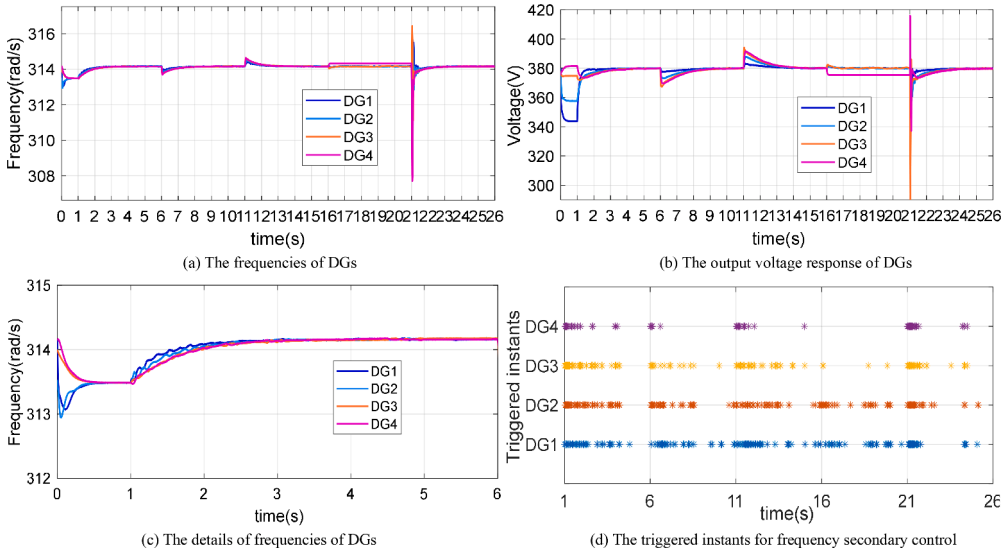

(a) The frequencies of DGs

(b) The output voltage response of DGs

(c) The details of frequency of DGs

(d) The triggered instants for frequency secondary control

**Fig. 8.** The simulation results under the dynamic event-triggered. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.).




(a) The frequencies of DGs

(b) The output voltage response of DGs

(c) The details of frequencies of DGs

(d) The triggered instants for frequency secondary control

**Fig. 9.** The simulation results under the proposed scheme.

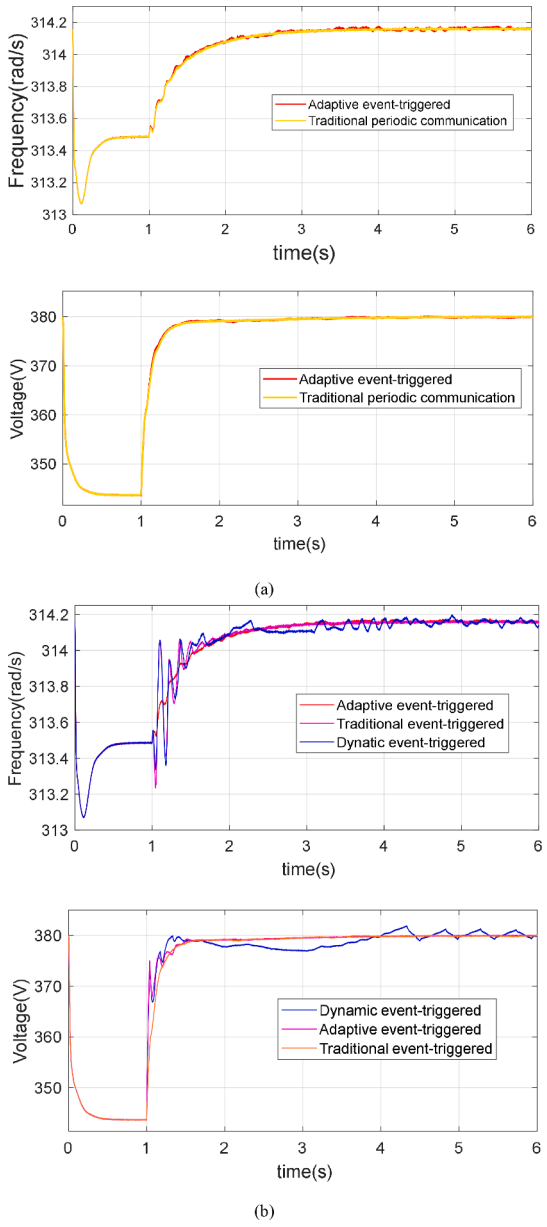



**Fig. 10.** Performance comparison under different schemes.

As shown in Fig. 7, traditional event-triggered can correspond to load changes and plug-and-play, ensuring that frequency and voltage converge to the reference value. However, by comparing Fig. 7 and 9, and as shown in performance comparison Fig. 10, static events-triggered significantly larger oscillations during the rise than adaptive event-triggered. As seen in Table I, the triggered instants of traditional event-triggered are significantly higher than that of the proposed scheme. The main reason is that the condition of $|e_{\omega i}(t)| \geq s_{\omega}$ is introduced into the trigger function of the proposed scheme, which effectively eliminates the frequent triggering of the system after stability.

As shown in Fig. 8 and 10 and Table I, dynamic event-triggered significantly reduces the triggering times due to the $\eta_i(t)$, but system performance deteriorates. In the plug-and-play scenario, the system diverges after DG4 is reconnected, so that the voltage and frequency cannot converge to the reference value.

As shown in Table I, the proposed scheme has the least time of communication. At the same time, as shown in Fig. 10(b), the proposed scheme has the smallest oscillation during the rising process, and the error after convergence is consistent with the traditional event-triggered. In Fig. 10(a), the frequency and voltage curve of the proposed scheme is consistent with that of traditional control in [16–19]. However, the communication burden of the proposed scheme is significantly reduced, and the system performance is maintained.

## 5. Conclusion

This article studies the problem of secondary control design in external interference and multiple time delays. It proposed an adaptive event-triggered $H_{\infty}$ consensus control strategy and evaluated the performance of the control in the load change and the plug-in-play. The simulation model is constructed and compared with other reported event-triggered strategies. The simulation results and comparisons indicate that this method has sufficient robustness for external interference and multiple delays, and the voltage and frequency can be restored to the reference value, which can effectively reduce the communication burden without affecting the system performance. It provides theoretical support for the design of secondary control for microgrids, which need to consider multiple delays and low bandwidth issues due to their establishment in remote and complex terrain. By the way, we will study economic problems in the tertiary control layers. And, improving the proposed methods for the problems of packet loss and spurious data injection in communication networks is also the subject of our future work.

## CRediT authorship contribution statement

**Chuang Liu:** Writing – original draft, Software. **Xinsheng Wang:** Writing – review & editing, Conceptualization. **Huaqiang Zhang:** Supervision, Methodology.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

No data was used for the research described in the article.

**Table 1**
Comparison of triggering times.

| DGs | Voltage | | | | Frequency | | | |
|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 1 | 2 | 3 | 4 |
| Traditional periodic communication | 5200 | 5200 | 5200 | 5200 | 5200 | 5200 | 5200 | 5200 |
| Traditional event-triggered | 950 | 2770 | 1392 | 323 | 1500 | 3213 | 2235 | 924 |
| Dynamic event-triggered | 602 | 1081 | 384 | 195 | 874 | 1220 | 488 | 428 |
| Adaptive event-triggered | 440 | 421 | 258 | 167 | 812 | 692 | 419 | 166 |

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